

Heaven's light is our guide



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Mechatronics Engineering

Lab Report

Experiment no.:09

Experiment name: Study On the Various Real-World Problem Solved Using Basic Logic in Ladder Diagram Programming.

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MTE 4102

Automation Sessional

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Experiment name: Study On the Various Real-World Problem Solved Using Basic Logic in Ladder Diagram Programming.

Objectives:

- To design PLC-based automation systems for real-life industrial processes such as fertilization, water tank filling, and packaging.
- To apply timers, counters, and sensor-based logic for process control, safety, and efficiency.
- To simulate automated production lines in OpenPLC for better understanding of sequential operations.

Theory

Programmable Logic Controllers (PLCs) are industrial computers designed for automation, timing, counting, and safety interlocks. They operate by reading inputs (such as sensors), processing them with logical instructions, and generating outputs (such as motors, pumps, or alarms).

- Timers are used for delays, monitoring process durations, and protection mechanisms.
- Counters track repeated events like products, bags, or boxes in a production line.
- Sensors provide feedback on machine or process status, allowing real-time decision-making.
- Interlocks and alarms improve safety by shutting down machines or alerting operators during faults.

These elements together allow PLCs to manage continuous, batch, and safety-critical processes, making them essential for modern industries.

Required Apparatus:

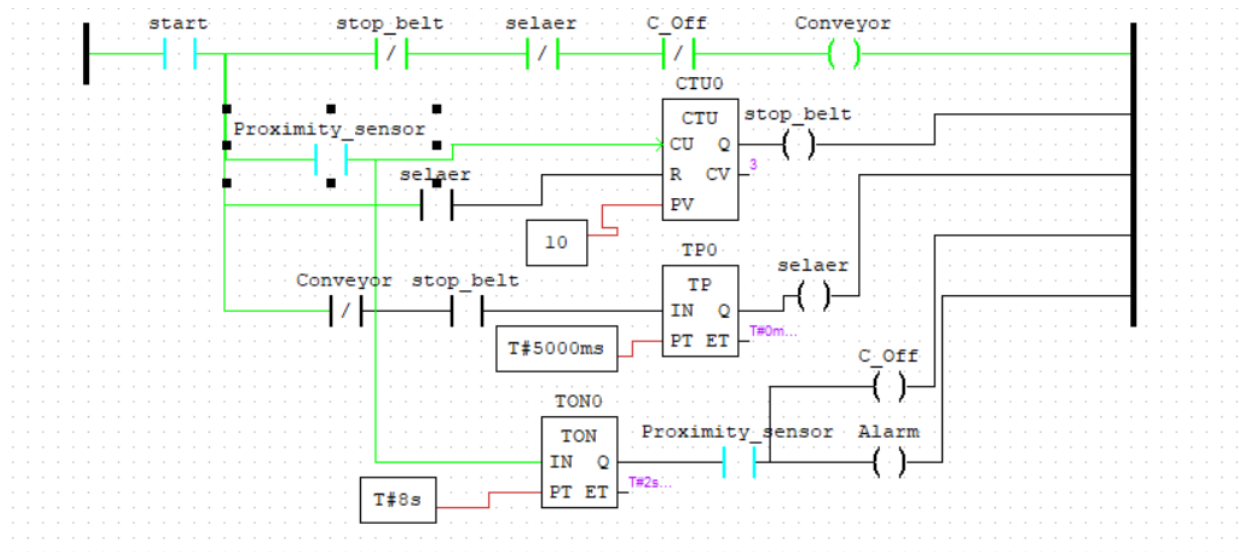
1. Laptop
2. OpenPLC Editor (PC Software)

Problem Statements & Ladder Diagram:

Problem1: Automated Fertilization System

A fertilizer bagging line drops material into bags on a conveyor. A proximity sensor detects each bag as it passes under the filling nozzle. The machine is designed to process 10 bags per minute. Count bags with a counter. When 10 bags are detected, stop the conveyor and run a sealer for 5 seconds (Timer). After sealing, reset the counter and restart the conveyor for the next batch. If no bag is detected within 8 seconds (Timer), assume a jam and stop the conveyor and trigger an alarm.

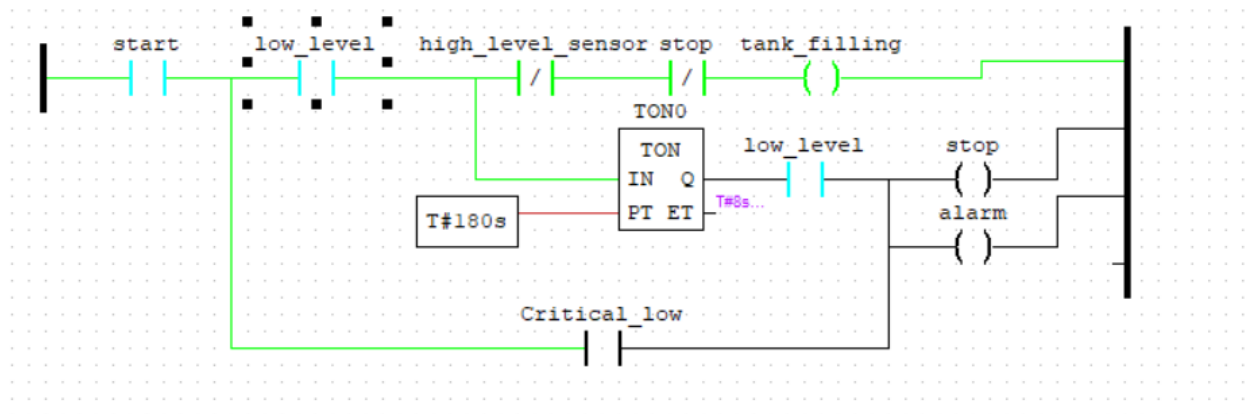
Ladder Diagram:



Problem2: Automatic Water Tank Filling System with Pump Protection

A factory uses a large water tank for cooling processes. The tank is equipped with three level sensors. When water drops to the Low-Level Sensor (L), the pump starts automatically to fill the tank. The pump keeps running until the High-Level Sensor (H) is reached, then it stops automatically. If the pump is running for more than 180 seconds continuously without reaching the high level, assume a leak or pump failure then emergency stops the pump and turn ON an alarm. If the Critical Low Sensor (CL) is triggered, stop the pump immediately and raise an alarm (dry-run protection).

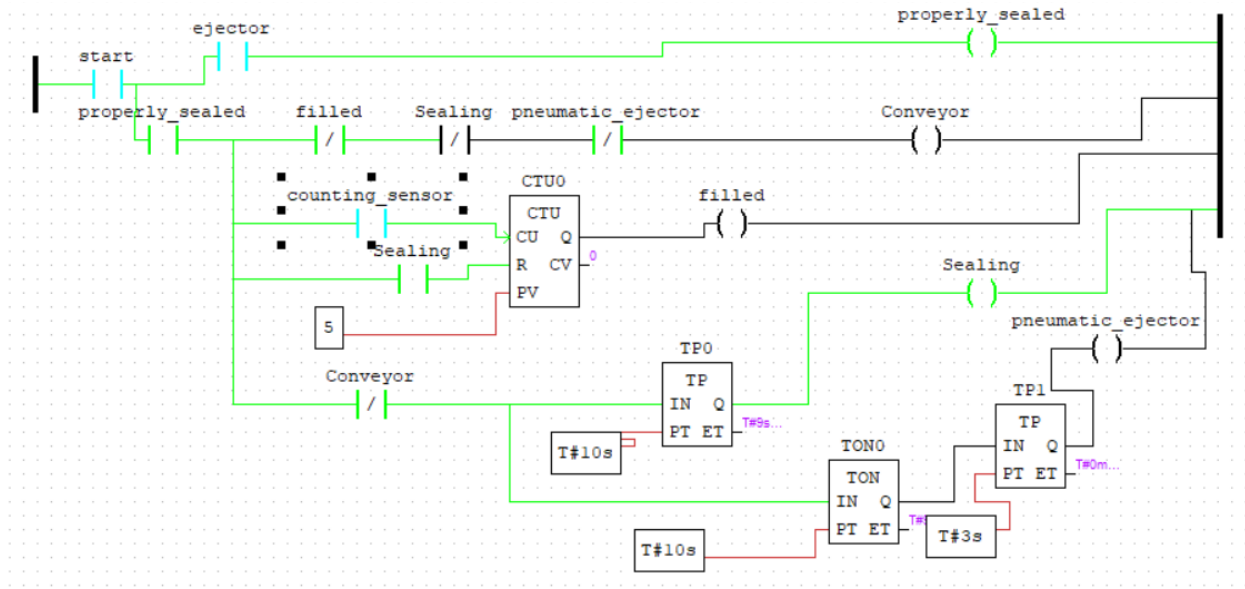
Ladder Diagram:



Problem3: Automatic Packaging Line

In a packaging plant, small boxes are packed into larger cartons. A carton ejector system ensures that every carton leaving the line is sealed properly. A sensor counts each small box passing on the conveyor. When 20 (taken 5 for simulation) boxes have been counted, the system, a carton is filled and the conveyor belt pauses. A sealing unit runs for 10 seconds to seal the carton. After sealing, a pneumatic ejector pushes the carton to a side conveyor. The ejector stays extended for 3 seconds (timer).

Ladder Diagram:



Discussion:

In the Automated Fertilization System, a proximity sensor detects each bag passing on the conveyor. A counter tracks the number of bags, and once 10 bags are counted, the conveyor stops while the sealer operates for 5 seconds. After sealing, the counter resets, and the cycle repeats. Additionally, if no bag is detected for 8 seconds, the system assumes a jam and triggers an alarm. This task demonstrates how PLCs manage batch processes and fault detection using counters and timers.

The Automatic Water Tank Filling System with Pump Protection applied three sensors for different water levels. The pump starts when the low-level sensor is triggered and stops when the high-level sensor is reached. A timer ensures the pump cannot run for more than 180 seconds continuously, preventing leaks or pump failure. The system also protects against dry-run damage by shutting down the pump and triggering an alarm if the critical low sensor is activated. This problem highlights the role of PLC in safety interlocks and preventive measures.

In the Automatic Packaging Line, a sensor counts small boxes on the conveyor. When 20 boxes are packed, the conveyor pauses, and the carton sealer operates for 10 seconds. Then, a pneumatic ejector pushes the sealed carton to a side conveyor for 3 seconds before resetting. This example reflects how PLCs coordinate sequential operations in a production line, combining counting, timing, and actuator control.

Together, these problems demonstrate how PLC programming integrates timers, counters, and safety logic to solve practical industrial challenges in material handling, fluid control, and packaging automation.

Conclusion:

This lab provided valuable insights into solving real-life industrial problems using PLC programming. The fertilization system showed how counters and timers manage batch processing while handling faults. The water tank system emphasized pump protection, sensor logic, and safety interlocks. The packaging line demonstrated sequential operations in automated production. Through these tasks, we strengthened our ability to design PLC-based solutions that ensure efficiency, reliability, and safety in industrial automation.